CUSTOM GRAPHICAL BINGO

User Manual

Bingo Card Generator + Bingo Moderator

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This manual describes the current Custom Graphical Bingo program family, including the printable card generator and the game moderator/projection program.

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1. Overview

Custom Graphical Bingo is a two-program system designed for Bingo games in which each Bingo number is represented by a custom graphic. The system separates card creation from game moderation so that the exact cards issued to players can be stored and later checked by the moderator.

Program	Primary purpose	Main result
Bingo Card Generator	Create unique printable Bingo cards.	BingoCards.pdf, BingoCards.xlsx, BingoCards.json
Bingo Moderator	Run the live game, call numbers, display graphics, and verify cards.	Live game + winner verification

The programs use the traditional 75-number Bingo distribution even when a custom five-letter header is used.

- B = 1–15
- I = 16–30
- N = 31–45
- G = 46–60
- O = 61–75

Important design principle

The graphic assigned to a number is controlled by its filename. For example, number 37 always uses Graphics\37.bmp. The same graphics folder should therefore be used by both programs.

2. How the Two Programs Work Together

The normal workflow is:

1. Prepare or update the 75 Bingo graphics.
2. Run the Bingo Card Generator.
3. Check that all 75 graphics are valid.
4. Optionally convert color graphics to greyscale.
5. Choose the header, card title, number of cards, FREE option, and winner-reduction option.
6. Generate the printable PDF cards.
7. Keep the generated BingoCards.json file with the cards.
8. Start the Bingo Moderator.
9. Load the same BingoCards.json and Graphics folder.
10. Select the winning pattern.
11. Open the projection window if a second display/projector is being used.
12. Call numbers during the game.
13. Check a player's card number when a Bingo is claimed.

BingoCards.json is the bridge between the two programs. It records the exact number layout of every generated card, allowing the moderator to verify a card by its card number.

3. Required Folder Structure

For the simplest setup, keep the programs and their supporting folders together:

```
Custom Bingo\
├── BingoCardGenerator_v6.exe or BingoCardGenerator_v6.py
├── BingoModerator_v22.exe or BingoModerator_v22.py
├── Graphics\
│   ├── 1.bmp
│   ├── 2.bmp
│   ├── ...
│   ├── 75.bmp
│   └── FREE.BMP   (when FREE is used)
└── Output\
    ├── BingoCards.pdf
    ├── BingoCards.xlsx
    ├── BingoCards.json
    ├── MasterGraphicReport.pdf
    └── BingoGameState.json
```

The compiled programs determine their working folder from the location of the EXE. This is important when moving the complete Bingo folder to another computer.

Do not rename the numbered graphics. The moderator expects 1.bmp through 75.bmp.

4. Bingo Number and Graphic Rules

Column	Numbers	Graphic filenames
B	1–15	1.bmp through 15.bmp
I	16–30	16.bmp through 30.bmp
N	31–45	31.bmp through 45.bmp
G	46–60	46.bmp through 60.bmp
O	61–75	61.bmp through 75.bmp

Every generated card contains five numbers from each traditional column range. The custom header is visual only; it does not change the number ranges.

If a FREE center square is enabled, the center position is represented internally as the FREE square and is displayed using Graphics\FREE.BMP.

5. Bingo Card Generator — Quick Start

Use the Card Generator whenever you need to create a new set of printable Bingo cards.

Step	Action
Step 1	Make sure Graphics contains 1.bmp through 75.bmp.
Step 2	Start Bingo Card Generator.
Step 3	Enter a five-letter header, such as BINGO, HOUSE, PARTY, or LUCKY.
Step 4	Enter the card title.
Step 5	Enter the number of cards to create.
Step 6	Choose whether to use the FREE center square.
Step 7	Choose whether to use Reduce Multiple Winners.
Step 8	Click CHECK GRAPHICS.
Step 9	If asked, choose whether color graphics should be converted to greyscale.
Step 10	Click MASTER GRAPHIC REPORT if you want a reference PDF of all 75 graphics.
Step 11	Click GENERATE PDF CARDS.
Step 12	Print BingoCards.pdf and retain BingoCards.json for the moderator.

6. Bingo Card Generator — Detailed Controls

Control	Purpose	Notes
5-letter header	Sets the five letters printed above the card.	Exactly five letters are required.
Card title	Title printed at the top of each card.	Defaults to Custom Bingo.
Number of cards	Number of unique cards to generate.	1 to 100,000 is available.
Use FREE center square	Makes the center square FREE.	Requires FREE.BMP.
Reduce Multiple Winners	Uses winner-aware generation intended to reduce simultaneous winners.	It is a generation option, not a guarantee of one winner.
Graphics folder	Location of 1.bmp–75.bmp and optional FREE.BMP.	Use Browse to select another folder.
CHECK GRAPHICS	Validates all 75 numbered BMP files.	Also offers optional greyscale conversion.
MASTER GRAPHIC REPORT	Creates a three-page reference PDF showing all 75 graphics.	Uses the same card-style layout.
GENERATE PDF CARDS	Creates the printable card set.	Also creates JSON and Excel output.

Header rules

The header must contain exactly five letters. The program normalizes the entry to uppercase and displays a live preview. The header does not alter the standard Bingo ranges.

Card title

The card title is printed above each generated card. If left blank, the program uses Custom Bingo.

Number of cards

The generator creates unique cards within the requested set. Card numbers are assigned sequentially starting at Card #0001.

FREE center

When enabled, the center square uses FREE.BMP. If FREE.BMP is missing, generation is stopped and the program reports the problem.

7. Checking and Converting Graphics to Greyscale

The CHECK GRAPHICS button is the recommended first step before generating cards.

14. Select the correct Graphics folder.
15. Click CHECK GRAPHICS.
16. The program verifies that all 75 numbered BMP files exist and can be opened.
17. If all 75 are valid, the program asks whether you want to convert color graphics to greyscale.
18. Choose Yes to convert color graphics in place, or No to leave them unchanged.

Only graphics identified as color are converted. Graphics already in greyscale are left alone. FREE.BMP is also converted when it exists and is in color.

Important: the greyscale conversion replaces the selected BMP files. Keep a backup of the original color graphics if you may want to restore them later.

8. Master Graphic Report

The Master Graphic Report is a printable reference showing all 75 Bingo graphics. It uses the same five-column card-style layout as the generated Bingo cards.

- The report is three pages.
- Page 1 contains graphics 1–25.
- Page 2 contains graphics 26–50.
- Page 3 contains graphics 51–75.
- Each cell shows the graphic number and the corresponding BMP image.
- The report is useful for checking the graphics before a game and for keeping a physical master reference.

The output is saved as:

Output\MasterGraphicReport.pdf

9. Generated Output Files

File	Purpose	Use
BingoCards.pdf	Printable Bingo cards.	Print and distribute to players.
BingoCards.xlsx	Spreadsheet representation of every generated card.	Review, archive, or inspect card layouts.
BingoCards.json	Machine-readable exact card definitions.	Load into Bingo Moderator for verification.
MasterGraphicReport.pdf	Reference report for graphics 1–75.	Inspect/print the graphic set.
BingoGameState.json	Current moderator game state.	Automatically used for game recovery.

10. Understanding BingoCards.json

BingoCards.json is not the printable card itself. It is the data file that tells the moderator exactly which numbers belong to each card.

It records information including:

- The custom five-letter header.
- The card title.
- Whether the FREE center is enabled.
- The FREE graphic filename when applicable.
- The number of cards generated.
- Each card number.
- The complete five-by-five number grid for every card.

Do not edit BingoCards.json manually unless you understand the card structure. The safest approach is to let the Card Generator create it.

11. Bingo Moderator — Quick Start

The Bingo Moderator is the live game-control program. It loads the card manifest, loads the graphics, randomly calls unused numbers from 1 through 75, displays the corresponding graphic, tracks the called numbers, and verifies cards.

19. Start Bingo Moderator.
20. Confirm that BingoCards.json is loaded.
21. Confirm that all 75 graphics are loaded.
22. Select the winning pattern.
23. Open the projection window if desired.
24. Choose animation and speed settings.
25. Click CALL NEXT NUMBER for each call.
26. When a player claims Bingo, enter their card number and check it.
27. Use COUNT WINNING CARDS when you want to see how many loaded cards currently satisfy the selected pattern.
28. Start a new clean game when beginning a completely new draw.

12. Bingo Moderator — Main Controls

Control	Purpose
LOAD CARD FILE	Select a BingoCards.json file.
LOAD GRAPHICS	Select the folder containing the numbered BMP graphics.
OPEN PROJECTION WINDOW	Open the separate player-facing display.
FULLSCREEN PROJECTION	Toggle the projection window between normal and fullscreen display.
Winning pattern	Select Line, X, L, O, U, T, or Full Page.
Animate reveal	Turns the call-reveal animation on or off.
Shuffle speed	Controls the speed of the graphic shuffle portion of the reveal.
Final zoom speed	Controls how quickly the selected final graphic grows from small to full size.
CALL NEXT NUMBER	Calls one unused number.

RECALL LAST	Returns to the most recently called number for review where provided in the main moderator controls.
CHECK CARD	Checks a specific card number against the current called numbers and selected pattern.
COUNT WINNING CARDS	Counts all loaded cards that currently satisfy the selected pattern.
NEW CLEAN GAME	Clears the current called-number history and starts a new draw.
Continue Same Card	Keeps the same cards and allows a new winning pattern to be played.

13. Calling Bingo Numbers

Each call is selected from the remaining unused numbers. A number is not called twice during the same game.

When CALL NEXT NUMBER is pressed:

- The program selects an unused number.
- The corresponding graphic is displayed.
- The B/I/N/G/O column letter is shown.
- The called-number board is updated.
- The called-number count is updated.
- The number becomes part of the saved game state.
- The last-four display is updated after the reveal completes.

The moderator can therefore be used as the authoritative source for the live draw rather than maintaining a separate manual list.

14. Call-Reveal Animation and Zoom Controls

The current moderator uses a two-part reveal when animation is enabled.

Setting	Effect
Animate reveal ON	Shows a visual reveal before publishing the final call.
Animate reveal OFF	The selected final number is displayed immediately.
Shuffle speed	Controls how quickly the teaser graphics cycle.
Final zoom speed	Controls the number of zoom steps and the time spent on each step.

The final graphic starts small and grows to full size. The final zoom-speed control changes the character of this effect rather than merely changing a small delay.

If animation is turned off while a reveal is in progress, the current selected number is completed immediately.

15. Projection Window

The Projection Window is intended for a second monitor, TV, or projector. It separates the player-facing display from the moderator's control screen.

The projection window contains:

- A large UNDER THE B/I/N/G/O indicator.

- The current Bingo graphic.
- A four-item last-called display.
- CALL NEXT NUMBER in the upper-left.
- CLOSE in the upper-left so the projection window can be closed directly.
- Fullscreen support.

The projection display is synchronized with the moderator's current call. During animation, the projection follows the reveal rather than replacing it with a static full-size image.

Recommended setup: place the moderator window on the operator's screen and move the projection window to the second display before enabling fullscreen.

16. Checking a Card and Verifying Winners

When a player claims Bingo, use the card number printed on the player's card.

29. Enter the card number in the card-check field.
30. Confirm that the correct winning pattern is selected.
31. Click CHECK CARD.
32. The moderator compares the exact stored card layout with the called-number history.
33. The result identifies whether that card currently satisfies the selected pattern.

When the card is a winner, the moderator can display the winning card preview with the winning cells identified.

COUNT WINNING CARDS checks every loaded card against the current called-number history. This is useful when the operator wants to know whether more than one generated card currently satisfies the selected pattern.

17. Bingo Winning Patterns

Pattern	Definition used by the moderator
Line	Any complete row, column, or diagonal.
X	Both diagonals.
L	A complete outside edge column plus the bottom row.
O	The outside perimeter.
U	Both outside side columns plus the bottom row.
T	The top row plus the center column.
Full Page	Every square is covered.

The exact pattern rules are built into the moderator. Always select the intended pattern before the game begins and before checking a claimed card.

18. New Game and Continue Same Cards

New Clean Game

Use NEW CLEAN GAME when you want to clear all called numbers and begin a completely new draw. The same generated cards can still be used, but the called-number history is reset.

Continue Same Card

Use Continue Same Card when the current cards remain valid but you want to play another pattern. The cards are retained, and the operator can select a new pattern.

For example, a game can use Line first and then continue with another pattern without generating a new set of cards.

19. Saved Game State and Recovery

The moderator automatically saves the current game state. The saved file is placed in the Output folder beside the card manifest.

The saved state includes:

- The loaded BingoCards.json path.
- The graphics folder.
- The custom header.
- The selected winning pattern.
- The numbers already called.
- The numbers remaining.
- The current number.
- The recall position.
- Animation enabled/disabled.
- Shuffle speed.
- Final zoom speed.

When the moderator starts and the matching BingoCards.json is found, it can offer to resume the saved game. This is particularly useful if the program or computer is restarted during a game.

The moderator saves state using an atomic replacement method so that normal game operation is not interrupted if state saving encounters a problem.

20. Recommended Game-Day Procedure

Before the event

- Back up the complete Bingo folder.
- Check all 75 graphics with CHECK GRAPHICS.
- Confirm the intended color/greyscale version of the graphics.
- Generate and inspect the Master Graphic Report.
- Generate the final Bingo cards.
- Open the BingoCards.pdf and confirm the cards look correct.
- Keep BingoCards.json with the final printed card set.
- Test the moderator with the same JSON and Graphics folder.
- Test the projection window and fullscreen mode.
- Test CALL NEXT NUMBER and CHECK CARD.

At the start of the game

- Start the moderator.
- Confirm the correct card file is loaded.
- Confirm all graphics are present.
- Select the winning pattern.
- Set animation and zoom preferences.
- Open and position the projection window.
- Start the game with CALL NEXT NUMBER.

When a player calls Bingo

- Pause calling.
- Record or read the player's card number.

- Enter the card number in the moderator.
- Click CHECK CARD.
- Confirm the reported pattern and winning cells.
- If needed, use COUNT WINNING CARDS to identify all current winners.

21. Troubleshooting

Problem	Likely cause	Solution
Graphics missing/invalid	One or more numbered BMPs is missing or unreadable.	Run CHECK GRAPHICS and correct the listed files.
FREE.BMP not found	FREE center is enabled but the graphic is missing.	Place FREE.BMP in the selected Graphics folder or disable FREE.
Cards cannot be generated	Graphics validation failed or the header/card count is invalid.	Correct the validation message and try again.
Moderator cannot load cards	Wrong BingoCards.json selected or file is invalid.	Load the BingoCards.json created by the Card Generator.
Moderator says graphics are missing	Graphics folder does not contain all 75 files.	Use LOAD GRAPHICS and select the correct folder.
Projection is blank	Projection window is open before a number is called.	Call a number or confirm the projection window is open correctly.
Projection will not close	Focus is on another window.	Use the CLOSE button in the projection window or close its window normally.
Animation is too fast/slow	Speed settings need adjustment.	Adjust Shuffle speed and Final zoom speed separately.
Game does not resume	Saved state belongs to a different card manifest or state is invalid.	Load the correct BingoCards.json and restart the moderator.
Cards appear different from the moderator	Different BingoCards.json files are being used.	Use the exact JSON generated with the printed cards.
Graphics look different from expected	The selected Graphics folder is different or was converted to greyscale.	Confirm the Graphics path and restore original graphics if necessary.

22. EXE / No-Python Deployment

The Bingo programs can be packaged as Windows EXE applications using PyInstaller. The target computer does not need Python installed when the compiled distribution is built correctly.

For a compiled deployment, keep the following beside the EXE:

- Graphics folder
- Output folder
- BingoCards.json when using the moderator

The compiled programs locate their working directory relative to the executable. Therefore, moving the complete application folder as a unit is preferable to moving only the EXE.

The EXE build should be tested on the actual computer that will run the game, particularly the graphics path, Output folder, projection display, and saved-state behavior.

23. Backup and File Management

For every production card set, keep an archive containing:

- The exact BingoCards.pdf that was printed.
- The matching BingoCards.json.
- The Graphics folder used to create the cards.
- The BingoCards.xlsx file, if you use it for records.
- The MasterGraphicReport.pdf.

Do not regenerate BingoCards.json from a different card set and expect it to validate an older printed card set. The JSON must correspond to the cards actually distributed.

If graphics are changed, treat the new graphics as a new production set. The number-to-graphic relationship is part of the visual identity of the cards and moderator display.

24. Technical Reference

Software requirements when running the Python versions

Component	Purpose
Python 3.10+	Runs the applications in source-code form.
Pillow	Loads, validates, converts, resizes, and displays BMP graphics.
ReportLab	Creates printable PDF cards and the Master Graphic Report.
openpyxl	Creates the BingoCards.xlsx workbook.

Core files

File	Role
BingoCardGenerator_v6(4)_MasterGraphicReport.py	Current generator feature set documented by this manual.
BingoModerator_v22(4)_CloseButton_WiderCalledNumbers.py	Current moderator feature set documented by this manual.
Graphics\1.bmp–75.bmp	Number graphics.
Graphics\FREE.BMP	Optional FREE center graphic.
Output\BingoCards.json	Exact card definitions used for verification.
Output\BingoGameState.json	Saved live-game state.

Design notes

- Cards are 5×5 and use traditional 75-number Bingo ranges.
- Generated cards are unique within the requested set.
- The moderator draws only unused numbers.
- The moderator checks the stored card layout rather than reconstructing a card from memory.
- Projection is a separate window so the operator can retain full control on the main screen.
- Game state is saved automatically.
- Animation affects presentation only; it does not change which number is selected.

Final checklist

☐ 75 graphics present and verified

☐ FREE.BMP present if FREE is enabled

☐ Header contains exactly five letters

☐ Correct number of cards selected

☐ Master Graphic Report reviewed

☐ BingoCards.pdf reviewed and printed

☐ Matching BingoCards.json retained

☐ Moderator loads the same JSON

☐ Moderator loads the same Graphics folder

☐ Winning pattern selected

☐ Projection tested

☐ Animation settings tested

☐ Backup completed

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